ForceRemove — User Guide

ForceRemove is a Niagara Program Object that bulk-removes components, points or extensions, including the ones Workbench won't delete the normal way. It backs up everything before deleting it, so the last run can be reversed.

Covers **N4 v3.3 / v3.4** and **AX v3.3-AX** (Niagara AX 3.5–3.8). The fields and actions are the same on both; see AX notes.

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1. How it works in one minute

```
you set:  operationMode + targetOrd (or targetList) + filters
            |
  Dry Run ──►  list what WOULD be deleted  ──►  log + results CSV   (no changes)
  Execute ──►  for each target:  back it up into ForceRemoveBackup
                                 then delete it                    ──►  log + results CSV
  Reverse ──►  put back everything the last Execute deleted, with its links
```

- It only runs when you press an action — nothing happens on its own.
- **Back up the station (.dist) and run Dry Run first, every time.**
- Every deleted item is copied into the `ForceRemoveBackup` folder first. If the copy fails, that item is **not** deleted.

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2. Every field, explained

Legend: **Set** = you set it · **Out** = the program writes it (read only).

About

Field		What it is
`version`	Out	The program version, e.g. `v3.4`.
`quickGuide`	Out	Short built-in help.

Last run

Field		What it is
`status`	Out	Live progress while it runs, then the summary: how many removed, skipped, failed, and where the log is. A refused path (outside the station folder) is reported here.

What to remove

Field		Default	What it is
`operationMode`	Set	BQL	Drop-down **Direct / BQL / CSV** — how the list of targets is built. See the three modes.
`targetOrd`	Set	blank	Direct: the component to remove (and everything matching inside it). BQL: the query. CSV: the CSV file. **Leave it blank in a template bog.**
`targetList`	Set	blank	Direct mode only: several targets, one ord per line; lines starting with `#` are ignored. Used instead of `targetOrd` when it isn't empty.
`nameFilter`	Set	blank	Only remove items whose name **contains** this text (case-sensitive, no wildcards). Blank (or `*`) = all.
`typeFilter`	Set	blank	Only remove items of exactly this type, `module:Type`, e.g. `control:NumericWritable` or `alarm:AlarmSourceExt`. Blank (or `*`) = all. Never applied to folders.

Ords can use names as Workbench shows them: `slot:/Drivers/BacnetNetwork/VAV X-01` is escaped for you (`VAV$20X$2d01`).

Files and logs

Field		Default (blank =)	What it is
`logFilePath`	Set	`file:^logs/ForceRemove.log`	Plain-text log of every step.
`resultsCsvPath`	Set	`file:^logs/ForceRemove_results.csv`	One row per item: what happened and why. Opens in Excel.

Field		Default (blank =)	What it is
`sampleCsvPath`	Set	`file:^logs/ForceRemove_SAMPLE.csv`	Where **Create Sample CSV** writes a starter CSV.
`maxArchives`	Set	10	How many old copies of the log and results CSV to keep. 0 = keep all.

All paths must be inside the station folder (`file:^…`); anything outside is refused.

Backups (reverse)

Field	What it is
`ForceRemoveBackup`	A folder inside the program holding copies of what the last Execute deleted. Emptied at the start of each Execute, so **Reverse only restores the most recent run**. Don't put anything else in it.

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3. The three modes

Mode	`targetOrd` holds	What gets removed
Direct	a component (or use `targetList` for several)	Everything inside it that matches the filters, **then the target itself**. Point it at one point to remove just that point.
BQL	a BQL query	Every row the query returns.
CSV	a CSV file	Every ord listed in the file.

Filters in each mode

- **Direct** — this is what the filters are for. Point at a folder and set `typeFilter` = `alarm:AlarmSourceExt` to remove only the alarm extensions under it.
- **BQL / CSV** — the query or file already chooses the targets, so leave both filters blank. A filter here acts as a safety net: a row that doesn't match is **SKIPPED** (and listed), not removed.

Order: BQL and CSV remove the shallowest items first (a folder before what's inside it); items already removed with their folder are skipped quietly. Direct with no filters removes the target as one unit; with a filter it works from the bottom up. For deep folder cleanups, run Execute a second time to collapse folders that became empty.

CSV format — one column, header row `Ord`:

```
Ord
station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points/OldPoint
station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_102/points/OldPoint
```

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4. Actions

Right-click the program → **Actions**.

Action (slot)	What it does
Execute (`execute`)	Run for real: back up, then delete.
Dry Run (`dryRun`)	A preview: logs and lists every item as `DRYRUN` exactly as Execute would, but deletes nothing, backs up nothing and changes no links.
Reverse (`reverse`)	Put back everything the last Execute removed: copies each backup to where it was and remakes its links (under their original names). Links whose other end no longer exists are skipped and logged `REVERSE LINK FAILED`. History records that were archived are **not** restored.
Cancel (`cancel`)	Stop after the item currently being removed.
Create Sample CSV (`createSampleCsv`)	Write a starter one-column CSV to `sampleCsvPath`.
Prune Archives (`pruneArchives`)	Apply `maxArchives` to the old logs right now.

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5. What it never deletes

In every mode, Dry Run included, these are logged `SKIPPED (PROTECTED)` instead:

- the station root
- the Services tree
- the `/Drivers` container itself (its contents can be removed)
- this program and its `ForceRemoveBackup` folder
- the folder the program sits in, **and everything in that folder** (sibling programs)

If the program can't work out where it is, it deletes nothing.

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6. Files it writes

All under the station's `logs` folder unless you change the paths:

File	Contents
`ForceRemove.log`	Every step, readable.

File	Contents
`ForceRemove_results.csv`	Columns: Timestamp, TargetName, TargetSlotPath, ExtName, ExtType, Status, Message, Mode, DurationMs.
`ForceRemove_reverse_manifest.csv`	What was backed up and where it came from (used by Reverse).
`ForceRemove_reverse_links.csv`	The links each removed item had (used by Reverse).
`ForceRemove_<date>_<time>.log` etc.	Archived copies from earlier runs (`maxArchives` kept).

Lines also go to the Application Director (info / warning / severe).

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7. Complete setups

Placeholders throughout — use your own paths.

A. Remove one device's points folder (Direct)

Field	Value
operationMode	Direct
targetOrd	`station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points`
nameFilter / typeFilter	blank

Dry Run → check the results CSV → Execute. The folder and everything inside it goes in one unit, and is backed up in one copy.

B. Strip alarm extensions off the points under a folder, keep the points (Direct)

Field	Value
operationMode	Direct
targetOrd	`station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101/points`
typeFilter	`alarm:AlarmSourceExt`

Only the alarm extensions go; the points stay.

C. Several devices at once (Direct with targetList)

```
# targetList
station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_101
station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_102
# station:|slot:/Drivers/BacnetNetwork/BXXXXX_VAV_103   (skipped - commented out)
```

D. Remove every point named TEST* in the drivers (BQL)

Field	Value
operationMode	BQL
targetOrd	`station:|slot:/Drivers|bql:select * from control:ControlPoint where name like 'TEST*'`
filters	blank

Never use a whole-station query such as `select * from baja:Component` — Execute would hit every unprotected component.

E. Remove a list from a spreadsheet (CSV)

1. **Create Sample CSV**, open `logs/ForceRemove_SAMPLE.csv`, put one ord per row under `Ord`.
2. Save it, e.g. as `Csv/RemoveList.csv` in the station's Files.
3. `operationMode` = CSV, `targetOrd` = `file:^Csv/RemoveList.csv`, filters blank.
4. Dry Run → Execute.

F. Undo

Press **Reverse** before the next Execute (the next Execute empties the backup folder).

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8. Results CSV

Status	Meaning
`REMOVED`	Backed up and deleted.
`DRYRUN`	Dry Run: would be deleted.
`SKIPPED`	Not deleted — the Message says why: protected, doesn't match the filters, already gone with its folder, or **backup failed - delete skipped to avoid data loss**.
`ERROR`	The delete was tried and failed; Message has the reason.

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9. Good habits

- Back up the station, Dry Run, read the results CSV, **then** Execute.
- Keep `targetOrd` blank in the copy you keep as a template.
- Put ForceRemove in its own folder (everything beside it is protected).
- Reverse straight away if something went wrong — the backup only holds the last run.

AX notes

- Program name **ForceRemove4AX**, code `ForceRemove4AX_v3.3.java`; works on AX 3.5–3.8.
- Same slots and actions. No imports to add (the AX code uses full class names).
- Display names on the AX slot sheet: Dry Run - No Changes, Reverse Removal, Cancel Request, Remove Old Logs, Create Sample CSV; `maxArchives` is limited to 0–10.
- The AX version doesn't sort its own property sheet (that's N4 v3.4).